

Introduction to Biophysics II: Quantum physics & biology

Lecture 19: Quantum coherence in biology - 4

Reading:
Modern optical spectroscopy, chapter 10

1

Last time... **The density matrix approach**

Excitation on donor
Excitation on acceptor

$$\Psi_B = C_1\psi_1 + C_2\psi_2 = C_1\phi_{1b}\chi_{1b}\phi_{2a}\chi_{2a} + C_2\phi_{1a}\chi_{1a}\phi_{2b}\chi_{2b}$$

In general, multistate (Shrödinger representation):

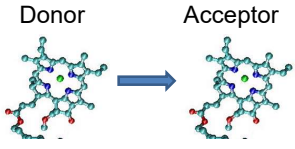
$$\begin{aligned} \Psi(t) &= c_1(t)\psi_1 + c_2(t)\psi_2 \\ c_n(t) &= C_n(t)\exp[-i(H_{nn}t/\hbar + \zeta_n)] \end{aligned}$$

Introduce **density matrix:**

$$\rho_{nm}(t) = c_n(t)c_m^*(t)$$

Expectation value for a measurement A:

$$A_{mn} = \langle \psi_m(\vec{r}) | \tilde{A} | \psi_n(\vec{r}) \rangle$$

$$\langle A(t) \rangle = \sum_n (\rho A)_{nn} = \text{Tr}(\rho \mathbf{A})$$


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Last time...

von Neumann & Liouville equations

Time-dependence of $\rho \rightarrow$ solve TD Schrödinger:

For the ensemble average of N systems

$$\rho_{nm}(t) = c_n(t)c_m^*(t)$$

von Neumann equation (1927)

$$\frac{\partial \bar{\rho}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, H]$$

Commutator:

$$[A, B] \equiv AB - BA$$

Stochastic Liouville Equation:

$$\frac{\partial \bar{\rho}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, H] + \mathbf{R} : \bar{\rho}$$

4th order tensor
2nd order tensor (matrix)

$$(\mathbf{R} : \bar{\rho})_{nm} = \sum_{i,k} R_{nmjk} \bar{\rho}_{jk}$$

$\mathbf{R}$ – relaxation matrix = set of rate constants for stochastic processes.

The double-dot product of a 4th-order tensor with a matrix results in a 2nd-order tensor

$$(\mathbf{R} : \bar{\rho})_{nm} \equiv \mathbf{R}_{nm} : \bar{\rho} \equiv \sum_{i,k} R_{nm,jk} \bar{\rho}_{jk} \left[= \text{Tr}(\mathbf{R}_{nm}^T \bar{\rho}) \right]$$

Frobenius inner product (or double-dot product): multiply matrices element-by-element and add up the resulting elements

For 4th order tensor and matrix: $\mathbf{R}:\bar{\rho}$ usually means double-dot product $\mathbf{R}:\bar{\rho}$

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Marcus & FRET revisited...

Marcus electron transfer and Förster energy transfer theories apply only to systems with weak intermolecular couplings, or in terms of the Liouville stochastic equation when the dephasing rate is fast compared to coupling:

$$|H_{21}|/\hbar \ll 1/T_2$$

Let us use the Liouville equation and explicitly introduce T_2 as decoherence time – i.e., oscillations in off-diagonal elements ρ are damped within T_2 time.

Solve the equation:

$$\frac{\partial \bar{\rho}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, H] + \mathbf{R} : \bar{\rho} \quad (\mathbf{R} \bar{\rho})_{nm} = \sum_{i,k} R_{nm,jk} \bar{\rho}_{jk}$$

FRET: 2-level system, state 1: excitation on donor, state 2 – excitation on acceptor

$$\frac{\partial}{\partial t} \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} = \frac{i}{\hbar} \left[\begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} - \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} \right] + \mathbf{R} \bar{\rho}$$

populations $\left\{ \begin{array}{l} \partial \rho_{11} / \partial t = (i/\hbar)(\rho_{12}H_{21} - \rho_{21}H_{12}) + \mathbf{R}_{11} : \bar{\rho} \\ \partial \rho_{22} / \partial t = (i/\hbar)(\rho_{21}H_{12} - \rho_{12}H_{21}) + \mathbf{R}_{22} : \bar{\rho} \end{array} \right. \begin{array}{l} \text{Assume } = 0 \\ \text{long decay} \end{array}$

Off-diagonal $\left\{ \begin{array}{l} \partial \rho_{12} / \partial t = (i/\hbar)[(\rho_{22} - \rho_{11})H_{21} + \rho_{21}(H_{11} - H_{22})] + \mathbf{R}_{12} : \bar{\rho} \\ \partial \rho_{21} / \partial t = (i/\hbar)[(\rho_{22} - \rho_{11})H_{21} + \rho_{21}(H_{11} - H_{22})] + \mathbf{R}_{21} : \bar{\rho} \end{array} \right. \begin{array}{l} \text{Assume} \\ 1/T_2 \text{ decay} \end{array}$

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Marcus & FRET revisited...

$$\frac{\partial \bar{\rho}_{nm}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, \mathbf{H}]_{nm} + (\mathbf{R}\bar{\rho})_{nm} \quad (\mathbf{R}\bar{\rho})_{nm} = \sum_{i,k} R_{nm,jk} \bar{\rho}_{jk} = \mathbf{R}_{nm} : \bar{\rho}$$

$$\frac{\partial}{\partial t} \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} = \frac{i}{\hbar} \left[\begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} - \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} \right] + \mathbf{R}\bar{\rho}$$

Consider ρ_{12} : $\partial \rho_{12} / \partial t = (i/\hbar) [(\rho_{22} - \rho_{11})H_{21} + \rho_{21}(H_{11} - H_{22})] + \mathbf{R}_{12} : \bar{\rho}$

Double-dot product

$$(\mathbf{R}_{12} : \bar{\rho}) = \begin{pmatrix} 0 & -1/T_2 \\ 0 & 0 \end{pmatrix} : \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} = -\rho_{12}/T_2$$

$$\partial \bar{\rho}_{12} / \partial t = (i/\hbar) \{ (\bar{\rho}_{11} - \bar{\rho}_{22})H_{12} + \bar{\rho}_{12}(H_{22} - H_{11}) \} - \bar{\rho}_{12}/T_2$$

Classical stochastic decay
of this off-diagonal element

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Marcus & FRET revisited...

We have:

$$0 = \partial \bar{\rho}_{12} / \partial t = (i/\hbar) \{ (\bar{\rho}_{11} - \bar{\rho}_{22})H_{12} + \bar{\rho}_{12}(H_{22} - H_{11}) \} - \bar{\rho}_{12}/T_2$$

If $|H_{21}|/\hbar \ll 1/T_2$, then $|\partial \bar{\rho}_{12} / \partial t|$ will be close to zero in the steady state,

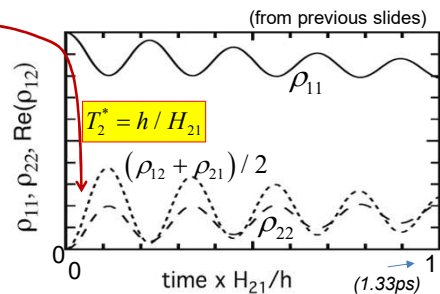
$\text{Re}(\rho_{12}) \sim \cos(t)$, slope is zero at $t \sim 0$

$$\partial \bar{\rho}_{12} / \partial t \sim 0$$

$$\bar{\rho}_{12} \approx \frac{(\bar{\rho}_{22} - \bar{\rho}_{11})H_{12}}{(H_{22} - H_{11}) + i\hbar/T_2}$$

Recall that: $\rho_{nm}(t) = c_n(t)c_m^*(t)$

$$\bar{\rho}_{21} = \bar{\rho}_{12}^* = \frac{(\bar{\rho}_{22} - \bar{\rho}_{11})H_{21}}{(H_{22} - H_{11}) - i\hbar/T_2}$$



(note: graph above is for slow $1/T_2$ rate to recall $\cos(t)$ shape of ρ_{12} . For faster dephasing you will not see oscillations at all)

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Marcus & FRET revisited...

$$\frac{\partial}{\partial t} \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} = \frac{i}{\hbar} \left[\begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} - \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} \right] + \mathbf{R}\bar{\rho}$$

$$\bar{\rho}_{12} \approx \frac{(\bar{\rho}_{22} - \bar{\rho}_{11})H_{12}}{(H_{22} - H_{11}) + i\hbar/T_2} \quad \bar{\rho}_{21} = \frac{(\bar{\rho}_{22} - \bar{\rho}_{11})H_{21}}{(H_{22} - H_{11}) - i\hbar/T_2}$$

Now find: $\partial \bar{\rho}_{22} / \partial t = (i/\hbar)(\bar{\rho}_{21}H_{12} - \bar{\rho}_{12}H_{21})$ $\mathbf{R}_{22ij}=0$,
(no decay for populations)

Substitute ρ_{12} and ρ_{21} :

$$\partial \bar{\rho}_{22} / \partial t = (\bar{\rho}_{11} - \bar{\rho}_{22}) |H_{21}|^2 \left(\frac{2\hbar/T_2}{(E_{12})^2 + (\hbar/T_2)^2} \right) \quad E_{12} = H_{22} - H_{11}$$

The steady-state rate of population of state 2 thus has a Lorentzian dependence on the energy gap E_{12} . If T_2^* is large, $T_2 = 2T_1$ and the Lorentzian has a shape of the respective absorption band expected for this lifetime.

The last term is simply Fermi's golden rule – the overlap of energies of donor and acceptor!

$$\partial \bar{\rho}_{22} / \partial t = (\bar{\rho}_{11} - \bar{\rho}_{22}) |H_{21}|^2 \frac{2\pi}{\hbar} \rho_s(0) \quad \text{Density of states}$$

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Absorption of weak light

Use the density matrix and the stochastic Liouville equation:

Given – a two-state system, states a and b (ground and excited)

Interaction with EM radiation at frequency $\omega = 2\pi\nu$ and amplitude E_0 only weakly dependent on time:

$$V_{ab} = -\boldsymbol{\mu}_{ab} \cdot \mathbf{E}(t) = -\boldsymbol{\mu}_{ab} \cdot [\mathbf{E}_0(t) \exp(i\omega t) + \mathbf{E}_0^*(t) \exp(-i\omega t)]$$

↑ Transition dipole moment, $\langle \psi_a | \hat{\boldsymbol{\mu}} | \psi_b \rangle$ Note: $V_{ab} = V_{ba}$

Assume the same μ direction and amplitude for all molecules in the ensemble.

Liouville equation, off-diagonal terms decay with T_2 :

$$\frac{\partial}{\partial t} \begin{pmatrix} \rho_{aa} & \rho_{ab} \\ \rho_{ba} & \rho_{bb} \end{pmatrix} = \frac{i}{\hbar} \left[\begin{pmatrix} \rho_{aa} & \rho_{ab} \\ \rho_{ba} & \rho_{bb} \end{pmatrix} \begin{pmatrix} E_a & V_{ab} \\ V_{ba} & E_b \end{pmatrix} - \begin{pmatrix} E_a & V_{ab} \\ V_{ba} & E_b \end{pmatrix} \begin{pmatrix} \rho_{aa} & \rho_{ab} \\ \rho_{ba} & \rho_{bb} \end{pmatrix} \right] + \mathbf{R}\bar{\rho}$$

$$\downarrow$$

$$\partial \bar{\rho}_{ab} / \partial t = (i/\hbar) \{ (\bar{\rho}_{bb} - \bar{\rho}_{aa}) \boldsymbol{\mu}_{ab} \cdot [\mathbf{E}_0 \exp(i\omega t) + \mathbf{E}_0^* \exp(-i\omega t)] + \bar{\rho}_{ab} E_{ba} \} - \bar{\rho}_{ab} / T_2$$

$$= (i/\hbar) (\bar{\rho}_{bb} - \bar{\rho}_{aa}) \boldsymbol{\mu}_{ab} \cdot [\mathbf{E}_0 \exp(i\omega t) + \mathbf{E}_0^* \exp(-i\omega t)] + (i\omega_{ba} - 1/T_2) \bar{\rho}_{ab},$$

Where $E_{ab} = E_b - E_a$, $\omega_{ba} = E_{ba}/\hbar$, and assume that ρ_{aa} and ρ_{bb} change slowly compared to ρ_{ab} , i.e., are almost constant under weak light

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Absorption of weak light

Have:

$$\partial \overline{\rho}_{ab} / \partial t = (i/\hbar)(\overline{\rho}_{bb} - \overline{\rho}_{aa}) \overline{\mu}_{ab} \cdot [\mathbf{E}_o \exp(i\omega t) + \mathbf{E}_o^* \exp(-i\omega t)] + (i\omega_{ba} - 1/T_2) \overline{\rho}_{ab},$$

Simplify in advance: the solution will combine terms $(\omega_{ba} \pm \omega)$ components, one will be much larger, and the second can be dropped. Assuming $E_{ab} = E_b - E_a > 0$:

$$\partial \overline{\rho}_{ab} / \partial t = (i/\hbar)(\overline{\rho}_{bb} - \overline{\rho}_{aa}) \overline{\mu}_{ab} \cdot \mathbf{E}_o \exp(i\omega t) + (i\omega_{ba} - 1/T_2) \overline{\rho}_{ab}$$

Since a periodic electric field drives the ensemble, the solution should also be periodic at the same frequency. Try this solution:

$$\overline{\rho}_{ab}(\omega, t) = \overline{\overline{\rho}}_{ab} \exp(i\omega t)$$

where $\overline{\overline{\rho}}_{ab}$ is an average of $\overline{\rho}_{ab}$ over an oscillation period of the field $(1/\omega)$

Substitute this solution into differential equation

$$\partial \overline{\rho}_{ab} / \partial t = \exp(i\omega t) \partial \overline{\overline{\rho}}_{ab} / \partial t + i\omega \cdot \exp(i\omega t) \overline{\overline{\rho}}_{ab} = 0 \text{ under steady state illumination}$$

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Absorption of weak light

$$\text{Have: } \partial \overline{\rho}_{ab} / \partial t = (i/\hbar)(\overline{\rho}_{bb} - \overline{\rho}_{aa}) \overline{\mu}_{ab} \cdot \mathbf{E}_o \exp(i\omega t) + (i\omega_{ba} - 1/T_2) \overline{\rho}_{ab}$$

$$\partial \overline{\rho}_{ab} / \partial t = i\omega \cdot \exp(i\omega t) \overline{\overline{\rho}}_{ab} \quad \overline{\rho}_{ab}(\omega, t) = \overline{\overline{\rho}}_{ab} \exp(i\omega t)$$

$$i\omega \cdot \exp(i\omega t) \overline{\overline{\rho}}_{ab} = (i/\hbar)(\overline{\rho}_{bb} - \overline{\rho}_{aa}) \overline{\mu}_{ab} \cdot \mathbf{E}_o \exp(i\omega t) + (i\omega_{ba} - 1/T_2) \overline{\overline{\rho}}_{ab} \exp(i\omega t)$$

and we solve for $\overline{\overline{\rho}}_{ab}$:

Similarly, solve or take *:

$$\overline{\rho}_{ab}(\omega, t) = \overline{\overline{\rho}}_{ab} \exp(i\omega t)$$

$$\overline{\overline{\rho}}_{ab} = \frac{(i/\hbar)(\overline{\rho}_{bb} - \overline{\rho}_{aa}) \overline{\mu}_{ab} \cdot \mathbf{E}_o}{i(\omega - \omega_{ba}) + 1/T_2}$$

$$\overline{\rho}_{ba} = \overline{\overline{\rho}}_{ba} \exp(-i\omega t)$$

$$\overline{\overline{\rho}}_{ba} = \frac{(i/\hbar)(\overline{\rho}_{bb} - \overline{\rho}_{aa}) \overline{\mu}_{ba} \cdot \mathbf{E}_o^*}{i(\omega - \omega_{ba}) - 1/T_2}$$

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Absorption of weak light

$$\overline{\rho_{ab}}(\omega, t) = \overline{\rho_{ab}} \exp(i\omega t)$$

$$\overline{\rho_{ab}} = \frac{(i/\hbar)(\overline{\rho_{bb}} - \overline{\rho_{aa}})\boldsymbol{\mu}_{ab} \cdot \mathbf{E}_o}{i(\omega - \omega_{ba}) + 1/T_2}$$

$$\overline{\rho_{ba}} = \overline{\rho_{ba}} \exp(-i\omega t)$$

$$\overline{\rho_{ba}} = \frac{(i/\hbar)(\overline{\rho_{bb}} - \overline{\rho_{aa}})\boldsymbol{\mu}_{ba} \cdot \mathbf{E}_o^*}{i(\omega - \omega_{ba}) - 1/T_2}$$

Can now find population of excited state:

$$\frac{\partial}{\partial t} \begin{pmatrix} \rho_{aa} & \rho_{ab} \\ \rho_{ba} & \rho_{bb} \end{pmatrix} = \frac{i}{\hbar} \left[\begin{pmatrix} \rho_{aa} & \rho_{ab} \\ \rho_{ba} & \rho_{bb} \end{pmatrix} \begin{pmatrix} E_a & V_{ab} \\ V_{ba} & E_b \end{pmatrix} - \begin{pmatrix} E_a & V_{ab} \\ V_{ba} & E_b \end{pmatrix} \begin{pmatrix} \rho_{aa} & \rho_{ab} \\ \rho_{ba} & \rho_{bb} \end{pmatrix} \right] + \mathbf{R}\overline{\rho}$$

$$\partial \overline{\rho_{bb}} / \partial t = (i/\hbar)(\overline{\rho_{ba}} V_{ab} - \overline{\rho_{ab}} V_{ba}) \quad \text{Assume long lifetime, } \mathbf{R}_{bb} = 0$$

$$\begin{aligned} \partial \overline{\rho_{bb}} / \partial t &\approx (i/\hbar) \{ [\exp(-i\omega t) \overline{\rho_{ba}}] [-\boldsymbol{\mu}_{ab} \cdot \mathbf{E}_o^* \exp(i\omega t)] \\ &\quad - [\exp(i\omega t) \overline{\rho_{ab}}] [-\boldsymbol{\mu}_{ba} \cdot \mathbf{E}_o \exp(-i\omega t)] \} \\ &= (\overline{\rho_{bb}} - \overline{\rho_{aa}}) \frac{|\boldsymbol{\mu}_{ba} \cdot \mathbf{E}_o|^2}{\hbar^2} \left(\frac{1}{i(\omega_{ba} + \omega) - 1/T_2} - \frac{1}{i(\omega_{ba} - \omega) + 1/T_2} \right) \\ &= (\overline{\rho_{aa}} - \overline{\rho_{bb}}) \frac{|\boldsymbol{\mu}_{ba} \cdot \mathbf{E}_o|^2}{\hbar^2} \left(\frac{2/T_2}{(\omega_{ba} - \omega)^2 + (1/T_2)^2} \right). \end{aligned}$$

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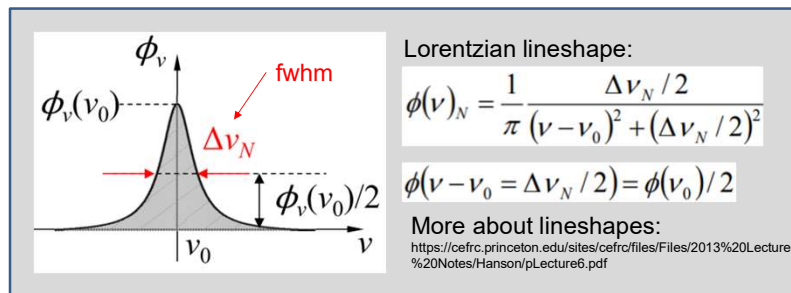
Absorption of weak light

The formation rate of the excited state:

$$\partial \overline{\rho_{bb}} / \partial t \approx (\overline{\rho_{aa}} - \overline{\rho_{bb}}) \frac{|\boldsymbol{\mu}_{ba} \cdot \mathbf{E}_o|^2}{\hbar^2} \left(\frac{2/T_2}{(\omega_{ba} - \omega)^2 + (1/T_2)^2} \right)$$

How does it relate to absorption?

How does the absorption spectrum look like?



Note: works for times $\gg T_2$. Assumes only one T_2 .

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Absorption of weak light

Formation rate of the excited state:

$$\partial \overline{\rho_{bb}} / \partial t \approx (\overline{\rho_{aa}} - \overline{\rho_{bb}}) \frac{|\boldsymbol{\mu}_{ba} \cdot \mathbf{E}_o|^2}{\hbar^2} \left(\frac{2/T_2}{(\omega_{ba} - \omega)^2 + (1/T_2)^2} \right)$$

Can rewrite in the form of Fermi's golden rule:

$$\partial \overline{\rho_{bb}} / \partial t = -\partial \overline{\rho_{aa}} / \partial t = (2\pi/\hbar) (\overline{\rho_{aa}} - \overline{\rho_{bb}}) |\boldsymbol{\mu}_{ba} \cdot \mathbf{E}_o|^2 \rho_s(E_{ba})$$

Spectral density
(density of states)

$\rho_s(E_{ba})dE$ = the number of states for which excitation energy is within dE of E_{ba}

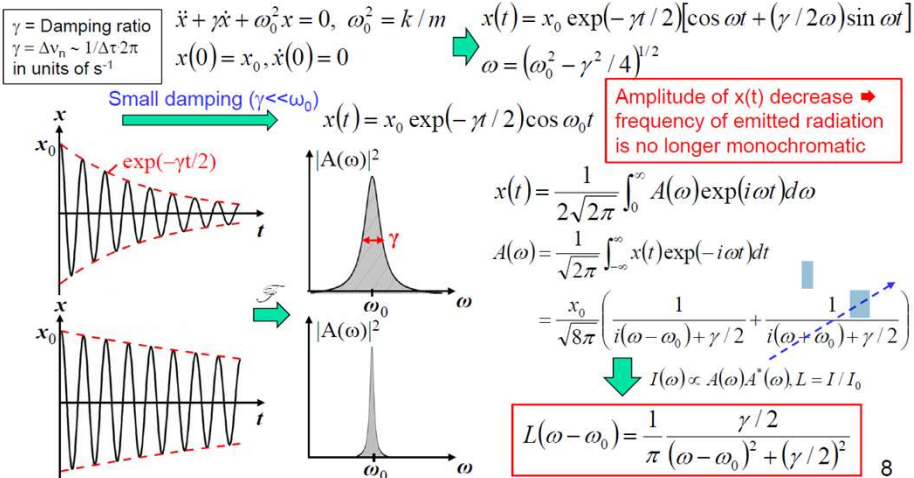
These equations also work for stimulated emission, except when dropping $\exp(-i\omega t)$ terms we should drop $\exp(+i\omega t)$ and vice versa (since $E_b < E_a$).

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Classical shape of a damped oscillator

Lorentzian shape from Fourier transform of damped oscillations:
"natural line broadening":

Lineshape derivation from damped oscillator model (Ref. Demtröder)



<https://cefc.princeton.edu/sites/cefc/files/Files/2013%20Lecture%20Notes/Hanson/pLecture6.pdf>

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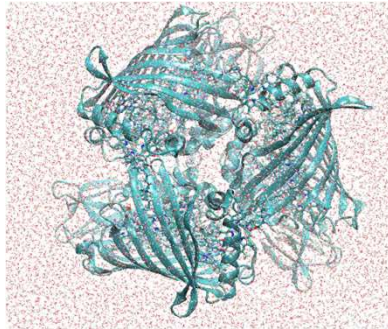
Relaxation matrix: dive deeper

$$\partial \bar{\rho} / \partial t = (i/\hbar) [\bar{\rho}, \mathbf{H}] + \mathbf{R} \bar{\rho} \quad \rho_{nm}(t) = c_n(t) c_m^*(t)$$

$$(\mathbf{R} \bar{\rho})_{nm} = \sum_{i,k} R_{nmjk} \bar{\rho}_{jk}$$

So far, the relaxation matrix contained phenomenological constants T_1 and T_2 .

We will be discussing elements of $\mathbf{R}$ in a more general way $\rightarrow$ dependence on interactions with surroundings



FMO-complex: thermal fluctuations

Write Hamiltonian as:

$$\mathbf{H}(t) = \mathbf{H}_0 + \mathbf{V}(t)$$

Independent
of time

interactions with
fluctuating electric or
magnetic fields from the
surroundings

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Relaxation matrix

$$\mathbf{H}(t) = \mathbf{H}_0 + \mathbf{V}(t)$$

Following work by:

Redfield, A.: *The theory of relaxation processes. Adv. Magn. Res. 1, 1–32 (1965)*

Redfield used the von Neumann-Liouville equation to find the effects of $\mathbf{V}(t)$ on ρ for the system:

$$\frac{\partial \bar{\rho}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, \mathbf{H}]$$

Note: Redfield used the interaction representation of the density matrix; we will use the Schrödinger representation

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Relaxation matrix

If we know $\rho(0)$ at time $t=0$, we can find $\rho(t)$ by integrating the von Neumann-Liouville equation:

$$\bar{\rho}(t) = \bar{\rho}(0) + \frac{i}{\hbar} \int_0^t [\bar{\rho}(0), \mathbf{H}(t_1)] dt_1$$

$$\mathbf{H}(t) = \mathbf{H}_0 + \mathbf{V}(t)$$

$$\frac{\partial \bar{\rho}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, \mathbf{H}]$$

However, using $\rho(0)$ under integral is an approximation as it should be changing. To improve accuracy, we can find ρ at intermediate time t_2 using the above equation, and then use $\rho(t_2)$ to calculate ρ at time t :

$$\bar{\rho}(t) = \frac{i}{\hbar} \int_0^t \left[\left(\bar{\rho}(0) + \frac{i}{\hbar} \int_0^{t_2} [\bar{\rho}(0), \mathbf{H}(t_1)] dt_1 \right), \mathbf{H}(t_2) \right] dt_2$$

Differentiate it over time:

$$\frac{\partial \bar{\rho}(t)}{\partial t} = \frac{i}{\hbar} [\bar{\rho}(0), \mathbf{H}(t)] + \left(\frac{i}{\hbar} \right)^2 \int_0^t [\bar{\rho}(0), \mathbf{H}(t_1), \mathbf{H}(t)] dt_1$$

Commutator in a commutator

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Relaxation matrix

Have:

$$\frac{\partial \bar{\rho}(t)}{\partial t} = \frac{i}{\hbar} [\bar{\rho}(0), \mathbf{H}(t)] + \left(\frac{i}{\hbar} \right)^2 \int_0^t [\bar{\rho}(0), \mathbf{H}(t_1), \mathbf{H}(t)] dt_1$$

$$\mathbf{H}(t) = \mathbf{H}_0 + \mathbf{V}(t)$$

$$\frac{\partial \bar{\rho}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, \mathbf{H}]$$

The ergodic hypothesis of statistical mechanics:

The ensemble average is equivalent to the average for a single system over time

→ With no loss of generality, assume that $\mathbf{V}(t)$ averages to 0 over time (can always adjust $\mathbf{H}_0$)

$$\overline{V_{nm}(t)} = \langle \psi_n | \tilde{\mathbf{V}}(t) | \psi_m \rangle = 0.$$



$$[\bar{\rho}(0), \mathbf{V}(t)] = 0$$

However: $(V_{nm}(t))^2$ or $V_{nm}(t)V_{jk}(t), \neq 0$

$$[\bar{\rho}(0), \mathbf{H}(t)] = [\bar{\rho}, \mathbf{H}_0]$$

Consider: $[\bar{\rho}(0), \mathbf{H}(t_1), \mathbf{H}(t)]$ density matrix at zero time

It includes sum of terms:

$$\bar{\rho}(0) V_{nm}(t_1) V_{jk}(t_2)$$

a correlation function or memory function

$$M_{nm, jk}(t_1, t_2) = \overline{V_{nm}(t_1) V_{jk}(t_2)}$$

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Correlation (Memory) function

$$\frac{\partial \overline{\rho}(t)}{\partial t} = \frac{i}{\hbar} [\overline{\rho(0), \mathbf{H}(t)}] + \left(\frac{i}{\hbar}\right)^2 \int_0^t [\overline{\rho(0), \mathbf{H}(t_1)}, \mathbf{H}(t)] dt_1 \quad \mathbf{H}(t) = \mathbf{H}_0 + \mathbf{V}(t)$$

Correlation (memory) function:

$$M_{nm, jk}(t_1, t_2) = \overline{V_{nm}(t_1) V_{jk}(t_2)}$$

= the ensemble average of the product of V_{nm} at time t_1 with V_{jk} at time t_2

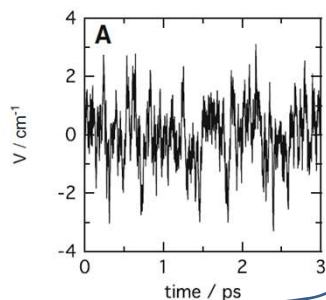
If V_{nm} and V_{jk} vary randomly, their average product should not depend on the particular times t_1 and t_2 , but only on the difference, $t = t_2 - t_1$.

Rewrite: $M_{nm, jk}(t) = \overline{V_{nm}(t_1) V_{jk}(t_1 + t)}$ ← average over t_1 and ensemble

Note: if $jk=nm$, this is autocorrelation function $a(t) = \int f(t_1) f(t_1 + t) dt_1$

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Correlation function: example

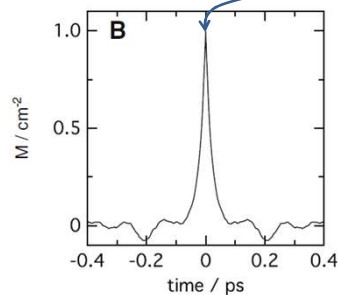


V(t)

Fluctuating interaction.

Root mean square 1 cm^{-1} , or:

$$M(0) = \int V(t_1) V(t_1 + 0) dt_1 = \int V^2(t_1) dt_1 \equiv \overline{V^2} = 1 \text{ cm}^{-2}$$



$M(t) =$

$$\int V(t_1) V(t_1 + t) dt_1$$

Autocorrelation function

Note: at longer times t autocorrelation $\rightarrow 0$

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Correlation function

Properties: $M_{nm,jk}(t) = \overline{V_{nm}(t_1)V_{jk}(t_1+t)}$ ← average over t_1 and ensemble

$$M_{nm,jk}(t) = \overline{\int V_{mn}(t_1)V_{jk}(t_1+t)dt_1}$$
 ← average over ensemble

At zero time, M is the mean product of the amplitudes

$$M_{nm,jk}(0) = \overline{\int V_{mn}(t_1)V_{jk}(t_1)dt_1} = \overline{V_{mn}} \cdot \overline{V_{jk}}$$

For “diagonal” terms with $jk=nm$ it is the mean square of the fluctuation:

$$M_{nm,nm}(0) = \overline{V_{nm}^2}$$

Note: at longer times t M goes to zero:

$$M_{nm,jk}(t \rightarrow \infty) = \overline{\int V_{mn}(t_1)V_{jk}(t_1+t)dt_1} = \overline{V_{mn}} \cdot \overline{V_{jk}} = 0$$

In the simplest model $M_{nm,jk}$ decays exponentially: $M_{nm,jk}(t) = M_{nm,jk}(0)\exp(-t/\tau_c)$

Correlation time: $\tau_c = \frac{1}{M_{nm,jk}(0)} \int_0^\infty M_{nm,jk}(t)dt$ $\int_0^\infty A_0 e^{-\tau_c t} dt = A_0 \tau_c$

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Spectral density function

The simplest model:

$$M_{nm,jk}(t) = M_{nm,jk}(0)\exp(-t/\tau_c) \quad \tau_c = \frac{1}{M_{nm,jk}(0)} \int_0^\infty M_{nm,jk}(t)dt$$

Note: Redfield analysis makes no other assumptions about the time course of M .

It assumes that correlations decay much faster than phenomenological times:

$$\tau_c \ll T_1, T_2^*$$

$M_{nm,jk}(t)$ describes the fluctuations of $V_{nm}V_{jk}$ as a function of time:

slows if fluctuations have long or short correlation times.

These fluctuations can also be described as a function of frequency (ν or $\omega=2\pi\nu$)

Spectral density function: $J_{nm,jk}(\omega) = \int_0^\infty M_{nm,jk}(t)\exp(i\omega t)dt$ (“Half” Fourier transform)

If M is symmetric, i.e. $M_{mn,jk}(t) = M_{nm,jk}(-t)$ and real:

$$J_{nm,jk}(\omega) = \frac{1}{2} \int_{-\infty}^\infty M_{nm,jk}(t)\exp(i\omega t)dt$$
 (Fourier transform)

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Relaxation matrix

$$R_{nm,jk} = \frac{1}{2\hbar^2} [J_{nj,mk}(\omega_{mk}) + J_{nj,mk}(\omega_{nj})] - \delta_{mk} \sum_i J_{ij,in}(\omega_{ij}) \\ - \delta_{nj} \sum_i J_{im,ik}(\omega_{ik}),$$

Problem: $R_{nn,mm} = R_{mm,nn}$, i.e. populations become equal in equilibrium!

For Boltzmann equilibrium must have $R_{mm,nn}/R_{nn,mm} = \bar{\rho}_{nn}/\bar{\rho}_{mm} = \exp(-\hbar\omega_{nm}/k_B T)$

The problem originates from ρ being only the density matrix of the system; it ignores energy exchange with the environment. Bath causes *only* dephasing, T_2^* .

Work-around by Redfield: multiply the spectral density function by the Boltzmann factor. More modern *semi-classical* (SC) approach: use Fermi-Dirac statistics:

$$J_{nm,jk}^{SC}(\omega_{\alpha\beta}) = J_{nm,jk}(\omega_{\alpha\beta}) [1 + \exp(\hbar\omega_{\alpha\beta}/k_B T)]^{-1}$$

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Relaxation matrix

$$J_{nm,jk}^{SC}(\omega_{\alpha\beta}) = J_{nm,jk}(\omega_{\alpha\beta}) [1 + \exp(\hbar\omega_{\alpha\beta}/k_B T)]^{-1}$$

The requirement for detailed balance:

$$J_{nn,mm}^{SC}(\omega_{nm}) = J_{mm,nn}^{SC} \exp(\hbar\omega_{nn}/k_B T)$$

Check:

$$J_{nn,mm}^{SC}(\omega_{nm}) \exp(\hbar\omega_{nn}/k_B T) = J_{nn,mm}(\omega_{nm}) \frac{\exp(\hbar\omega_{nn}/k_B T)}{1 + \exp(\hbar\omega_{nn}/k_B T)} \\ = J_{nn,mm}(\omega_{nm}) [1 + \exp(-\hbar\omega_{nn}/k_B T)]^{-1} \\ = J_{mm,nn}(\omega_{nn}) [1 + \exp(\hbar\omega_{nn}/k_B T)]^{-1} = J_{mm,nn}^{SC}(\omega_{nn}).$$

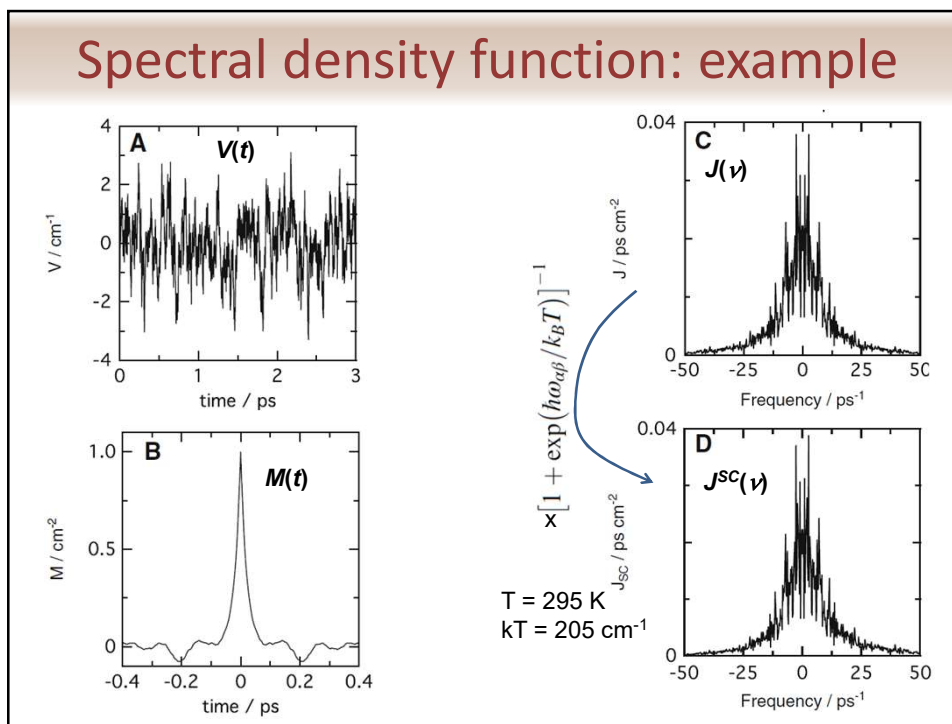
Note: semiclassical spectral density function J can be written as a sum of two components, even and odd functions of ω

→ Fourier transform of J^{SC} is thus complex.

The imaginary part can be viewed as carrying directionality of time.

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Spectral density function: example



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Relaxation matrix

With J^{SC} we have relaxation matrix:

$$R_{nm, jk} = \frac{1}{2\hbar^2} \left(\frac{J_{nj, mk}(\omega_{mk})}{1 + \exp(\hbar \omega_{mk} / k_B T)} + \frac{J_{nj, mk}(\omega_{nj})}{1 + \exp(\hbar \omega_{nj} / k_B T)} \right) - \delta_{mk} \sum_i \frac{J_{ij, in}(\omega_{ij})}{1 + \exp(\hbar \omega_{ij} / k_B T)} - \delta_{nj} \sum_i \frac{J_{im, ik}(\omega_{ik})}{1 + \exp(\hbar \omega_{ik} / k_B T)}$$

Note: delta functions mean that the last two terms are not zero *only* when indices $m=k$ (i.e., for $R_{nm, jm}$) or $n=j$ ($R_{nm, nk}$), or both for $R_{nm, nm}$.

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Relaxation matrix example

For a 2-level system

$$\frac{\partial \bar{\rho}_{nm}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, \mathbf{H}]_{nm} + \sum_{i,k} R_{nm,ik} \bar{\rho}_{ik}$$

$$R_{11,11} = -\frac{1}{\hbar^2} \left(\frac{J_{21,21}(\omega_{21})}{1 + \exp(\hbar\omega_{21}/k_B T)} \right) \quad R_{22,22} = -\frac{1}{\hbar^2} \left(\frac{J_{12,12}(\omega_{12})}{1 + \exp(\hbar\omega_{12}/k_B T)} \right)$$

$$R_{11,22} = -\frac{1}{\hbar^2} \left(\frac{J_{12,12}(\omega_{12})}{1 + \exp(\hbar\omega_{12}/k_B T)} \right) \quad R_{22,11} = -\frac{1}{\hbar^2} \left(\frac{J_{21,21}(\omega_{21})}{1 + \exp(\hbar\omega_{21}/k_B T)} \right)$$

$$R_{12,12} = \frac{-1}{2\hbar^2} [J_{11,11}(0) - 2J_{11,22}(0) + J_{22,22}(0)] + \frac{1}{2} [R_{22,11} + R_{11,22}]$$

$$R_{21,21} = \frac{-1}{2\hbar^2} [J_{22,22}(0) - 2J_{22,11}(0) + J_{11,11}(0)] + \frac{1}{2} [R_{11,22} + R_{22,11}]$$

$$R_{12,21} = \frac{1}{\hbar^2} J_{12,21}(0)$$

$$R_{12,21} = \frac{1}{\hbar^2} J_{21,12}(0)$$

Note: if V_{12} and V_{21} are real:

$$J_{21,12}(\omega) = J_{12,21}(\omega)$$

$$J_{21,21}(\omega) = J_{12,12}(\omega)$$

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Relaxation matrix: meaning

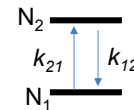
$$\frac{\partial \bar{\rho}_{nm}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, \mathbf{H}]_{nm} + \sum_{i,k} R_{nm,ik} \bar{\rho}_{ik}$$

For a 2-level system:

$$\frac{\partial \bar{\rho}_{11}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, \mathbf{H}]_{11} + R_{11,11} \bar{\rho}_{11} + R_{11,22} \bar{\rho}_{22} + R_{11,12} \bar{\rho}_{12} + R_{11,21} \bar{\rho}_{21}$$

Compare semiclassical part:

$$\frac{dN_1}{dt} = R_{11,11} N_1 + R_{11,22} N_2$$



$k_{21} = -R_{11,11}$ - rate of energy transfer from state 1 (state 1 decay)

$k_{12} = R_{11,22}$ - rate of energy transfer from 2 to 1

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Relaxation matrix example

For 2-level system

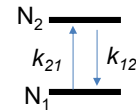
$$R_{11,11} = -\frac{1}{\hbar^2} \left(\frac{J_{21,21}(\omega_{21})}{1 + \exp(\hbar\omega_{21}/k_B T)} \right)$$

Population of state 1:

$$\partial \bar{\rho} / \partial t = (i/\hbar) [\bar{\rho}, \mathbf{H}] + \mathbf{R} \bar{\rho}$$

$$\partial \bar{\rho}_{11} / \partial t = (i/\hbar) [\bar{\rho}, \mathbf{H}]_{11} + R_{11,11} \bar{\rho}_{11} + R_{11,22} \bar{\rho}_{22}$$

$$\text{Semiclassical part: } \partial N_1 / \partial t \sim R_{11,11} N_1 + R_{11,22} N_2$$



$$k_{21} = -R_{11,11} = \hbar^{-2} J_{21,21}(\omega_{21}) [1 + \exp(\hbar\omega_{21}/k_B T)]^{-1} = \hbar^{-2} J_{21,21}^{SC}(\omega_{21})$$

Similarly, population of state 2:

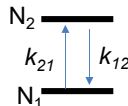
$$\partial \bar{\rho}_{22} / \partial t = (i/\hbar) [\bar{\rho}, \mathbf{H}]_{22} + R_{22,11} \bar{\rho}_{11} + R_{22,22} \bar{\rho}_{22}$$

$$k_{12} = -R_{22,22} = \hbar^{-2} J_{12,12}(\omega_{12}) [1 + \exp(\hbar\omega_{12}/k_B T)]^{-1} = \hbar^{-2} J_{12,12}^{SC}(\omega_{12})$$

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Relaxation matrix example

For 2-level system



$$k_{21} = -R_{11,11} = \hbar^{-2} J_{21,21}(\omega_{21}) [1 + \exp(\hbar\omega_{21}/k_B T)]^{-1} = \hbar^{-2} J_{21,21}^{SC}(\omega_{21})$$

$$k_{12} = -R_{22,22} = \hbar^{-2} J_{12,12}(\omega_{12}) [1 + \exp(\hbar\omega_{12}/k_B T)]^{-1} = \hbar^{-2} J_{12,12}^{SC}(\omega_{12})$$

The transfer rate between states 1 and 2 depends on the spectral density of EM fields from the surrounding that fluctuate at ω_{21} (energy difference E_{21})

$$\text{Recall: } J_{12,12} \leftarrow M_{12,12} \leftarrow \text{autocorrelation of } V_{12} \equiv \langle \psi_1 | V(t) | \psi_2 \rangle$$

Note: this looks like spectral overlap! The levels are "broadened" by bath.

$$\text{Also, we have Boltzmann equilibrium: } k_{21}/k_{12} = \exp(\hbar\omega_{12}) = \exp[(H_{11} - H_{22})/k_B T]$$

$$\text{The equilibration time constant: } 1/T_1 = k_{21} + k_{12} = \hbar^{-2} [J_{21,21}^{SC}(\omega_{21}) + J_{12,12}^{SC}(\omega_{12})]$$

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Relaxation matrix example

For 2-level system

Consider off-diagonal term: $\partial \bar{\rho}_{12} / \partial t = (i / \hbar) [\bar{\rho}, \mathbf{H}]_{12} + R_{12,12} \bar{\rho}_{12} + \dots$ ↓ Dephasing rate

$$R_{12,12} = \frac{-1}{2\hbar^2} [J_{11,11}(0) - 2J_{11,22}(0) + J_{22,22}(0)] + \boxed{\frac{1}{2}[R_{22,11} + R_{11,22}]}$$

Note that the integral J is = amplitude*width of the M distribution: ↑ = 1/2T₁

$$J_{nm,jk}(\omega) = \int_{-\infty}^{\infty} M_{nm,jk}(t) \exp(i\omega t) dt$$

$$J_{nm,jk}(\omega = 0) = \int_{-\infty}^{\infty} M_{nm,jk}(t) dt = \tau_c M_{nm,jk}(t = 0)$$

Also recall: $M_{nm,jk}(0) = \overline{V_{nm} V_{jk}}$

$$\begin{aligned} [J_{11,11}(0) - 2J_{11,22}(0) + J_{22,22}(0)] &= \tau_c (\overline{V_{11} V_{11}} - 2\overline{V_{11} V_{22}} + \overline{V_{22} V_{22}}) \\ &= \tau_c (\overline{V_{11} - V_{22}})^2. \end{aligned}$$

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Relaxation matrix example

For 2-level system

Consider off-diagonal term: $\partial \bar{\rho}_{12} / \partial t = (i / \hbar) [\bar{\rho}, \mathbf{H}]_{12} + R_{12,12} \bar{\rho}_{12} + \dots$ ↓ Dephasing rate

$$R_{12,12} = \frac{-1}{2\hbar^2} [J_{11,11}(0) - 2J_{11,22}(0) + J_{22,22}(0)] + \boxed{\frac{1}{2}[R_{22,11} + R_{11,22}]}$$

$$\Rightarrow R_{12,12} = \frac{-1}{2\hbar^2} \tau_c (\overline{V_{11} - V_{22}})^2 + 1/2T_1$$
↑ = 1/2T₁

Recall *transverse relaxation time* $1/T_2 \approx 1/T_2^* + 1/2T_1$

↓ *Pure dephasing time:*

$$\frac{1}{T_2^*} = \frac{\tau_c}{2\hbar^2} (\overline{V_{11} - V_{22}})^2$$

Note: mean value of $\langle V_{11} - V_{22} \rangle = 0$ ($V(t)$ fluctuates around zero by definition)

$$(\overline{V_{11} - V_{22}})^2 \leftarrow \text{Variance of energy distribution}$$

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Relaxation matrix

For 2-level system

$$R_{12,12} = \frac{-1}{2\hbar^2} [J_{11,11}(0) - 2J_{11,22}(0) + J_{22,22}(0)] + \frac{1}{2} [R_{22,11} + R_{11,22}]$$

$\downarrow$ $\uparrow = -1/T_2^*$ $\uparrow = 1/2T_1$

$$1/T_2 \approx 1/T_2^* + 1/2T_1$$

For n-level system: (more terms for $1/2T_1$)

$$\frac{1}{T_2} = \frac{1}{T_2^*} - \frac{1}{2} \sum_{j \neq n, m} (R_{nm, jj} + R_{mm, jj})$$

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Relaxation matrix

Consider the Liouville stochastic equation: $\partial \bar{\rho} / \partial t = (i/\hbar) [\bar{\rho}, \mathbf{H}] + \mathbf{R} \bar{\rho}$

$$\frac{\partial \bar{\rho}_{nm}}{\partial t} = \frac{i}{\hbar} [\bar{\rho}, \mathbf{H}]_{nm} + \sum_{i,k} R_{nm, ik} \bar{\rho}_{ik}$$

For each equation $\partial \bar{\rho}_{nm} / \partial t$ we have the sum of products $R_{nm, ik} \bar{\rho}_{ik}$ over all other elements. Can be *very large!*

However, for an off-diagonal element $\bar{\rho}_{nm}$ ($n \neq m$) the effect of another diagonal element $\bar{\rho}_{jk}$ ($j \neq k$) is greatest when $\omega_{nm} \approx \omega_{jk}$. $\leftarrow$ then the two matrix elements oscillate in synch and $R_{nm, ik} \bar{\rho}_{ik}$ keeps consistently increasing (or decreasing) the derivative $\partial \bar{\rho}_{nm} / \partial t$. When out of resonance, this term averages to zero.

Rotating wave approximation: keep only $R_{nm, ik}$ that are resonant, $\omega_{nm} \approx \omega_{jk}$.

For a 3- or more level system (see Redfield paper for details):

Additional terms of the relaxation matrix for the transfer of coherence between two pairs of states.

These terms allow one off-diagonal density matrix element ($\bar{\rho}_{jk}$) to increase at the expense of another ($\bar{\rho}_{nm}$). Again, these terms are most important if $\omega_{jk} \approx \omega_{nm}$.

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Relaxation matrix: 3 states

What we found: fluctuations of the surroundings at angular frequency ω_{21} can cause transitions between states that differ in energy by $\hbar\omega_{21}$

Suppose: correlation function for these fluctuations

$M_{21,21}(t)$ is an exponential

Classical density function $J_{21,21}(\omega)$ is then Lorentzian $\rightarrow$

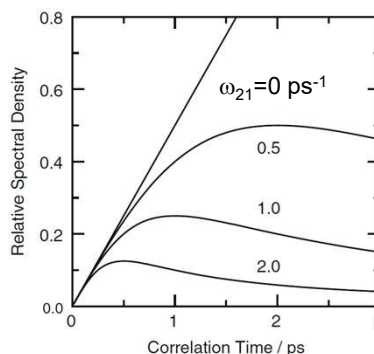
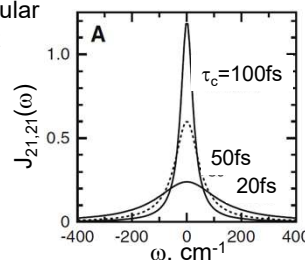
Width $\sim 1/\tau_c$.

Faster dephasing $\rightarrow$ higher intensity, small ω and lower intensity at large ω

What matters is $J_{21,21}(\omega)$ value at $1 \rightarrow 2$ transition energy, i.e. at $\omega = \omega_{21}$

Plot value of $J(\omega)$ at several values of ω as a function of τ_c (i.e., Lorentzian width $1/\tau_c$) $\rightarrow$

Spectral density at ω_{21} peaks when $\tau_c = 1/\omega_{12}$
Speeding up fluctuations increases the rate of transfer $1 \rightarrow 2$ as long as $\tau_c > 1/\omega_{21}$, but decrease after that



Used Fig. 2.12A from book, decay times are wrong in fig, corrected here

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Redfield's theory limitation

Redfield's theory assumptions:

- Fluctuations of surroundings is fast compared to T_1 and T_2^*
- Applies only to times $t \gg \tau_c$

Additional reading: more general Relaxation function, section 10.7 in the textbook

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EXTRA: Relaxation matrix

Revisit the starting equation GOT BORED, SKIP IN 2021...:

$$\mathbf{H}(t) = \mathbf{H}_0 + \mathbf{V}(t)$$

$$\frac{\partial \overline{\rho}(t)}{\partial t} = \frac{i}{\hbar} [\overline{\rho}(0), \mathbf{H}(t)] + \left(\frac{i}{\hbar}\right)^2 \int_0^t [\overline{\rho}(0), \mathbf{H}(t_1)], \mathbf{H}(t) dt_1$$

$$\downarrow [\overline{\rho}(0), \mathbf{H}(t)] = [\overline{\rho}(0), \mathbf{H}_0] + [\overline{\rho}(0), \mathbf{V}(t)] = [\overline{\rho}(0), \mathbf{H}_0]$$

$$[[\rho_0, \mathbf{H}(t_1)], \mathbf{H}(t)] = [\rho_0 \mathbf{H}(t_1) - \mathbf{H}(t_1) \rho_0, \mathbf{H}(t)] =$$

$$= \rho_0 \mathbf{H}(t_1) \mathbf{H}(t) - \mathbf{H}(t_1) \rho_0 \mathbf{H}(t) - \mathbf{H}(t) \rho_0 \mathbf{H}(t_1) + \mathbf{H}(t) \mathbf{H}(t_1) \rho_0$$

$$\rho_0 \mathbf{H}(t_1) \mathbf{H}(t) = \rho_0 \{ \mathbf{H}_0 + \mathbf{V}(t_1) \} \{ \mathbf{H}_0 + \mathbf{V}(t) \} =$$

$$= \rho_0 \mathbf{H}_0 \mathbf{H}_0 + \rho_0 \mathbf{H}_0 \mathbf{V}(t) + \rho_0 \mathbf{H}_0 \mathbf{V}(t_1) + \rho_0 \mathbf{V}(t_1) \mathbf{V}(t)$$

$$\mathbf{H}(t_1) \rho_0 \mathbf{H}(t) = \{ \mathbf{H}_0 + \mathbf{V}(t_1) \} \rho_0 \{ \mathbf{H}_0 + \mathbf{V}(t) \} =$$

$$= \mathbf{H}_0 \rho_0 \mathbf{H}_0 + \mathbf{H}_0 \rho_0 \mathbf{V}(t) + \mathbf{H}_0 \rho_0 \mathbf{V}(t_1) + \mathbf{V}(t_1) \rho_0 \mathbf{V}(t)$$

$$\overline{\rho}(0) \overline{V_{nm}(t_1) V_{jk}(t_2)},$$